PROJECT-BASED LEARNING

PHASE-II EVALUATION

FOOD RESCUE

Smart Food Rescue and Redistribution System

Team ID: DSCPP-III-2026-T381

Department of Computer Science & Engineering
Graphic Era (Deemed to be University), Dehradun
Academic Session 2026–27

Team & Mentor Details

Team Information

Team ID: DSCPP-III-2026-T381
Semester: 3rd
Section: ML2, ML3, DS1, ML4
Domain: Social Impact

Team Members

1. Aditya Dutta (Lead) – 2510373905
2. Utsav Bajoria – 2510370581
3. Vishwajeet Singh – 2510380298
4. Raghav Pandey – 2510380113

Mentor

Mr. Chirag Tyagi

Interactions completed: 8

Brief Discussion: Introduce the team, project domain, mentor, and responsibilities of the team members.

Phase-I Recap & Feedback Incorporated

Phase-I Status

Literature/background study on food-rescue platforms completed; requirements and user roles finalized; system architecture and class diagram prepared.

Feedback Received

- Improve donation prioritization.
- Improve coordination between NGOs and donors.
- Make delivery planning more efficient.

Changes Incorporated

- Design priority selection based on expiry and urgency.
- Plan requirement-based donation matching.
- Include distance-based route planning.

Brief Discussion: Explain the major feedback received during Phase-I and clearly demonstrate how it has been

Problem Statement & Finalized Objectives

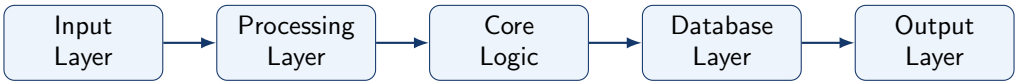
Problem Statement

Restaurants, hostels, households, and events often generate surplus food that is still safe to consume, while NGOs and people in need may face shortages. Without proper coordination, usable food goes to waste. Food Rescue connects donors with NGOs and records food quantity, location, type, and expiry, prioritizing donations by urgency and distance.

Finalized Objectives

1. Reduce food wastage by connecting food donors with NGOs and volunteers.
2. Match surplus food with NGO requirements by food type, quantity, and location.
3. Prioritize eligible donations based on expiry time and urgency.
4. Support efficient pickup and delivery route planning using graph algorithms.

Updated System Architecture / Workflow



Major Components

- UI layer for donors, NGOs, and volunteers.
- Logic engine for constraint checking and donation allocation.
- Record management for donations, users, and deliveries.

Changes Since Phase-I

- Defined a modular, layered architecture.
- Added priority-based matching and route-planning logic.
- Connected donor, NGO, and volunteer workflows.

Brief Discussion: Present the updated architecture and explain how individual modules communicate with each other.

Module Design & Functional Requirements

Major Modules

1. Core DSA, Integration & UI
2. Volunteer & Delivery Module
3. NGO & Requirement Module
4. Donor & Food Donation Module

Functional Requirements

- Web front-end (HTML/CSS/JavaScript) for three user roles.
- Priority-based donation selection and matching.
- Dijkstra-based delivery route planning.
- Delivery tracking and file-based record storage.

Module Dependency / Data Flow



Technology Stack & Development Methodology

Technology Stack

- Programming: C++ (STL, OOP, DSA), HTML, CSS, JavaScript
- Data storage: File-based CSV / text records
- Library: Standard Template Library (STL)
- Development tool: VS Code / Code::Blocks
- Version control: Git / GitHub

Development Methodology

1. Requirement Analysis
2. System Design
3. Module Development
4. Integration
5. Testing & Validation

Brief Discussion: Explain why the selected technologies are appropriate and how the team is following the

Implementation of PBL Course Concepts

Course	Concepts Actually Implemented	Project Module
Data Structures in C	Priority queue for urgent donations; queue for NGO requests; hash table for donor/NGO/volunteer lookup; graph and Dijkstra's algorithm for delivery routes.	DSA-based matching and optimization
OOPs with C++	Classes for Donor, FoodDonation, NGO, Volunteer, Requirement, and Delivery; STL containers and file handling. Include inheritance/polymorphism only if implemented.	Application and data-management layer

Important: Mention only concepts actually implemented in the project.

Be prepared to explain where and how each concept is used in the code.

Development & Implementation Progress

Module	Status	Work Completed
Core DSA, Integration & UI	in-progress	Implement core data structures and coordinate module integration.
Volunteer & Delivery	Completed	Implement core data structures and coordinate module integration.
NGO & Requirement	Completed	Develop NGO requirement matching and validate matching cases.
Donor & Food Donation	Completed	Complete donor records, integration, testing, and documentation as needed.

Overall Project Completion: *Enter actual percentage based on completed work*

Brief Discussion: Clearly differentiate between completed, in-progress and pending development. Do not present proposed work as completed work.

Working Prototype / Demonstration

Implemented Features

- Food donation entry and record management (if working).
- NGO requirement registration and matching (if working).
- Priority-based donation selection (if working).
- Volunteer assignment / route planning (if working).

Prototype / Output



Testing, Results & Validation

Test / Module	Expected Result	Evidence in Test Code	Status
Donor / volunteer	Registered records display.	Calls to `displayDonor()` and `displayAll()`.	Code present
NGO matching	Rice donation (20 units) checked against 20-unit requirement.	Calls to `matchWithDonations()` and pending/fulfilled displays.	Verify output
Priority queue	One queued task is processed.	Three tasks enqueued; `dequeue()` processes one.	Verify output
Volunteer assignment	Volunteer assigned to intended delivery.	Creates delivery ID 401 but searches for ID 501.	Fix ID mismatch

Results / Validation: Run tests and record console output before marking Pass. Fix ID mismatch: delivery 401 is created, but 501 is searched.

GitHub Repository & Development Activity

Repository Information

- GitHub Link: Insert actual repository URL
- README.md: Updated
- Contributors: Aditya Dutta , Utsav Bajoria , Vishwajeet Singh , Raghav Pandey
- Repository Status: Updated
- Branches: Main

Development Activity

- Commit history maintained by each member.
- Use descriptive commit messages.
- Individual work visible in commit history.
- Upload latest tested implementation.
- Maintain README and setup instructions.

Requirement: All team members must be contributors and must **commit individually using their own GitHub accounts**.

Brief Discussion: Show repository activity, README, contributor list and commit history during the

Individual Contribution

Member	Role	Technical Contribution	GitHub Commits
Aditya Dutta	Core DSA, Integration & UI	Core data structures, integration, and UI work actually completed.	15
Utsav Bajoria	Volunteer & Delivery	Volunteer records, pickup assignment, and delivery tracking, where implemented.	7
Vishwajeet Singh	NGO & Requirement	NGO requirement records and donation-matching work, where implemented.	6
Raghav Pandey	Donor & Food Donation	Donor records and food donation entry, where implemented.	7

Individual Evaluation

Every member should be prepared to explain and demonstrate their own technical contribution to the project.

Challenges, Remaining Work & Phase-III Roadmap

Challenges Faced

- Ensuring suitable donation-to-NGO matching.
- Coordinating interfaces and records across modules.
- Maintaining correct priority and delivery records.

Solutions / Learning

- Use modular design, suitable data structures, and test cases.
- Apply DSA and OOP concepts to a practical social-impact problem.

Phase-III Roadmap

1. Complete remaining module implementations.
2. Integrate donor, NGO, volunteer, and core logic.
3. Test edge cases and optimize matching and routes.
4. Demonstrate and validate the final prototype.
5. Complete README, test evidence, and final report.

Expected Outcomes & References

Expected Final Outcomes

- Integrated working food-rescue prototype.
- Validated donation matching and route-planning results.
- Potential use by donors, NGOs, and volunteers.
- Future extension to live maps and persistent databases.
- Potential for further evaluation and improvement.

References

1. Cormen et al., Introduction to Algorithms.
2. Balagurusamy, Object-Oriented Programming with C++.
3. C++ Standard Library documentation.
4. Course lecture notes on Data Structures and OOP.
5. Add the actual project GitHub repository URL.